ORSI - gyakorlat

Feladatmegoldás

Definíció

Az S program egy adott $K \in inv_S(\bigwedge_{Q \in Init_h} Q)$ invariáns mellett megfelel a következő kikötéseknek

$P \rhd_h R$ -nek	ha $P \wedge K \rhd_S R \wedge K$
$P \mapsto_h R$ -nek	ha $P \wedge K \mapsto_S R \wedge K$
$P \hookrightarrow_h R$ -nek	ha $P \wedge K \hookrightarrow_S R \wedge K$
$P \in inv_h$ -nak	ha $P \wedge K \in inv_S(\bigwedge_{Q \in Init_h} Q)$
$P \in FP_h$ -nak $(FP_h \Rightarrow P$ -nek$)$	ha $\varphi_S \wedge K \Rightarrow P$
$P \in TERM_h$ -nak $(P \hookrightarrow FP_h$ -nak$)$	ha $sp(s_0, P) \wedge K \in TERM_S$ $(sp(s_0, P) \wedge K \hookrightarrow_S \varphi_S)$

Variáns függvény tétel.

P, Q feltételek, $t : \mathbb{A} \to \mathbb{Z}$ variáns függvény.

$$\frac{P \wedge \neg Q \Rightarrow t > 0, \quad \forall m \in N : P \wedge \neg Q \wedge t = m \hookrightarrow_S (P \wedge t < m) \vee Q}{P \hookrightarrow_S Q}$$

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$P \in TERM_h$ **bizonyítás esetén:**
Legyen $K \in inv_S(\bigwedge_{Q \in Init_h} Q)$,

$$\frac{\neg\varphi_S \wedge K \Rightarrow t > 0, \quad \forall t' \in N : \neg\varphi_S \wedge K \wedge t = t' \hookrightarrow_S (K \wedge t < t') \vee \varphi_S}{S \text{ megfelel tetszőleges } P \text{ feltétel esetén a } P \in TERM_h \text{ feltételnek.}}$$

1.Feladat

Megfelel-e az alábbi S program a megadott specifikációnak?

$V = vektor\big([1..n], \mathbb{N}_0\big)$ $\qquad$ $\mathbf{A} = \underset{x}{V} \times \underset{m}{\mathcal{Z}}$ $\qquad$ $\mathbf{B} = \underset{x'}{V}$

$$S \;=\; \left(m := 0,\ \left\{ \mathop{\square}_{i=1}^{n} \ x_i,\ m := x_i - 1,\ m + 1,\ \text{ha } x_i > 0 \right\} \right)$$

1. $Q = \big(\forall k \in [1..n] : x_k = x'_k\big) \in INIT_h$
2. $P = \Big(m + \sum_{k=1}^{n} x_k = \sum_{k=1}^{n} x'_k\Big) \in inv_h$
3. $R = \Big(m = \sum_{k=1}^{n} x'_k\Big) \in FP_h$
4. Mutass variánsfüggvényt, amivel $Q \in TERM_h$ bizonyítható lenne.

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Bizonyítás:

Kell $K \in inv_S(Q)$, ami mellett megfelel. Tegyük fel, hogy $K = IGAZ$.
(Ha nem jó, kereshetünk másik invariánst...)

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$P \in inv_S(Q)$, azaz
$\left(m + \sum_{k=1}^{n} x_k = \sum_{k=1}^{n} x'_k\right) \in inv_S\left(\forall k \in [1..n] : x_k = x'_k\right)$

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$P \in inv_S(Q)$,

- $Q \Rightarrow lf(s_0, P)$, azaz

  $$\left(\forall k \in [1..n] : x_k = x'_k\right) \Rightarrow lf(s_0, \left(m + \sum_{k=1}^{n} x_k = \sum_{k=1}^{n} x'_k\right))$$

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$\neg\varphi_S = \neg\Big(\mathop{\wedge}_{i=1}^{n} (x_i = 0)\Big)$

$$V = \mathit{vektor}\Big([1..n], \mathbb{N}_0\Big) \qquad \mathbf{A} = \underset{x}{V} \times \underset{m}{\mathcal{Z}} \qquad \mathbf{B} = \underset{x'}{V}$$

$$S \;=\; \left(m := 0,\; \left\{ \mathop{\square}_{i=1}^{n} x_i,\; m := x_i - 1,\; m + 1,\; \text{ha } x_i > 0 \right\} \right)$$

4. Mutass variánsfüggvényt, amivel $Q \in \mathit{TERM}_h$ bizonyítható lenne.

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Legyen $t = \sum_{k=1}^{n} x_k$, $\neg\varphi_S \Rightarrow t > 0$, illetve
a program egyetlen utasítása sem növeli, és $\neg\varphi_S$ esetén van olyan, amelyik csökkenti.

2.Feladat

Megfelel-e az alábbi S program a megadott specifikációnak?

$V = vektor\big([1..n], \mathbb{Z}\big)$ $\qquad$ $\mathbf{A}{=}\underset{x}{V}\times\underset{s}{\mathbb{Z}}$ $\qquad$ $\mathbf{B}{=}\underset{x'}{V}$

$$S \;=\; \left(s := 0,\ \left\{ \mathop{\square}_{i=1}^{n} \ s,\ x_i := s+1,\ x_i+1,\ \text{ha } 2|x_i \right\} \right)$$

1. $Q = (\forall i \in [1..n] : x_i = x_i') \in INIT_h$
2. $P = \Big(s = \sum_{i=1}^{n} \chi(2|x_i') - \sum_{i=1}^{n} \chi(2|x_i)\Big) \in inv_h$
3. $R = \Big(s = \sum_{i=1}^{n} \chi(2|x_i')\Big) \in FP_h$
4. Mutass variánsfüggvényt, melynek segítségével a $Q \in TERM_h$ kikötés bizonyítható!

Megjegyzés:

- $\chi(igaz) = 1,\ \chi(hamis) = 0$
- $2|x$ jelentse, hogy kettő osztója az x-nek, azaz x páros.

$V = vektor\Big([1..n], \mathbb{Z}\Big)$ $\qquad$ $\mathbf{A}{=}V \times \mathbb{Z}$ $\qquad$ $\mathbf{B}{=}V$

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2. Tegyük fel, hogy $K = IGAZ$

Kell: $P \in inv_S(Q)$

- $Q \Rightarrow If(s_0, P)$

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Teljesül! $(x = x')$

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2. Tegyük fel, hogy $K = IGAZ$

Kell: $P \in inv_S(Q)$

- $P \Rightarrow If(S, P)$

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  $= (2|x_k \rightarrow P^{s \leftarrow s+1, x_k \leftarrow x_k+1}) \wedge (2 \not| x_k \rightarrow P)$

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  $= (2|x_k \to P^{s \leftarrow s+1, x_k \leftarrow x_k+1}) \wedge (2 \not| x_k \to P)$

  $= (2|x_k \to \Big(s+1 = \sum_{i=1}^{n} \chi(2|x_i') - \sum_{\substack{i=1 \\ i \neq k}}^{n} \chi(2|x_i) - \chi(2|x_k+1)\Big) \wedge (2 \not| x_k \to P)$

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$$\varphi_S = \bigwedge_{i=1}^{n} \Big(2 \not| x_i \vee (s = s+1 \wedge x_i = x_i+1)\Big) = \bigwedge_{i=1}^{n} (2 \not| x_i \vee FALSE) = \bigwedge_{i=1}^{n} (2 \not| x_i)$$

Legyen $K = P$, beláttuk $P \in inv_S(Q)$.

$$\varphi_S \wedge K = \bigwedge_{i=1}^{n} (2 \not| x_i) \wedge \Big(s = \sum_{i=1}^{n} \chi(2|x_i') - \sum_{i=1}^{n} \chi(2|x_i)\Big)$$

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Legyen $t = n - s + 1$, illetve legyen itt is $K = P$.

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$K \Rightarrow s \leq n \Rightarrow t > 0$, illetve

$V = vektor\Big([1..n], \mathbb{Z}\Big)$ $\qquad$ $\mathbf{A}{=}V \times \mathbb{Z}$ $\qquad$ $\mathbf{B}{=}V$

$x \quad s \qquad\qquad x'$

$$S = \left(s := 0, \ \left\{ \mathop{\square}_{i=1}^{n} \ s, \ x_i := s+1, \ x_i + 1, \ \text{ha } 2|x_i \right\} \right)$$

$$P = \left(s = \sum_{i=1}^{n} \chi(2|x_i') - \sum_{i=1}^{n} \chi(2|x_i) \right)$$

4. Mutass variánsfüggvényt, amivel $Q \in TERM_h$ bizonyítható lenne.

Kell:

$\neg\varphi_S \wedge K \Rightarrow t > 0$, illetve

$\forall t' \in N : \neg\varphi_S \wedge K \wedge t = t' \hookrightarrow_S (K \wedge t < t') \vee \varphi_S$

$\neg\varphi_S = \neg \bigwedge_{i=1}^{n} (2 \not| x_i) = \bigvee_{i=1}^{n} (2|x_i)$

Legyen $t = n - s + 1$, illetve legyen itt is $K = P$.

$K \Rightarrow s \leq n \Rightarrow t > 0$, illetve
a program egyetlen utasítása sem növeli, és $\neg\varphi_S$ esetén van olyan, amelyik csökkenti.